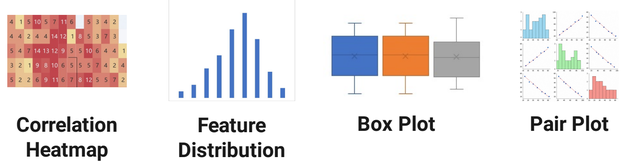
1 Data Acquisation

- Apose-IoT-23
- BoT-IoT
- CIC-IoMT-2024
- CIC-IoT-2025

2 Preprocessing & Feature Harmonization

- Label Unification (Benign / DDoS)
- Feature Alignment (Unified Feature Set)
- Missing Value Handling (Imputation)
- Encoding (if needed)
- Normalization (Min-Max / Z-score)

Data Analysis & Visualization



3 Temporal Sequence Contruction (Sliding Window)

Unified Feature Vector

$x_t \in \mathbb{R}^d$

f_1 f_2 f_3 ... f_d

Sliding Window of size T

$t - T + 1$: x_{t-T+1} x_{t-T+2} ... x_t

$t - T + 2$: x_{t-T+2} x_{t-T+3} ... x_{t+1}

t : x_t x_{t+1} ... x_{t+T-1}

Sequence Input

$x_i \in \mathbb{R}^{T \times d}$

4 Hybrid Feature Extraction (Dual Branch)

CNN Branch (Spatial Feature Learning)

1D Convolution + ReLU → Batch Normalization → Max Pooling → Flatten + Dense → f_{cnn}

BiLSTM Branch (Temporal Feature Learning)

h_1 h_2 ... h_T

x_1 x_2 ... x_T

Concat → f_{lstm}

5 Attention-Based Feature Fusion

Concatenation

$f = f_{cnn} \oplus f_{lstm}$

Attention Mechanism

Weighted Feature

$f' = \alpha \odot f$

Attention Computation

$$e_i = tanh\,(W_a f + b_a)$$

$$\alpha_i = \frac{exp\,(e_i)}{\sum_{j=1}^{d} exp\,(e_j)}$$

$$f' = \alpha \odot f$$

6 Classification & Output

- Dense Layer + ReLU
- Dropout
- Output Layer (Softmax)

Prediction

- Benign (0)
- DDoS (1)

7 Training & Evaluation

Train / Test Split (Intra / Cross-Dataset) → Model Training (Adam Optimizer)

Lost Function (Binary Cross-Entropy)

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic}\,log(\hat{y}_{ic})$$

Performance Evaluation (Accuracy, Precision, Recall, F1-score, ROC-AUC)

8 Explainability (XAI)

- SHAP
- LIME
- ANOVA